

AI-powered bioinformatic analysis using R

Instructor: Wei-Hua Chen (HUST, China) **License:** CC BY-NC 4.0 **Course Repository:** <https://github.com/evolgeniusteam/R-for-bioinformatics>

Course Overview

16 lectures, divided into two parts: - **Part 1 (Talks 1–6):** R Programming Foundations - **Part 2 (Talks 7–16):** Data Analysis & Multi-omics Applications

Part 1: R Programming Foundations

Talk	Title	Source (merged from)
01	Course Overview & AI-assisted Setup	talk00 + talk01 + talk00_github
02	R Language Basics I	talk02
03	R Language Basics II	talk03
04	Data Wrangling with tidyverse	talk04 + talk05
05	Data Visualization with ggplot2	talk06 + talk09

Talk	Title
07	Statistical Analysis & Machine Learning Basics
08	Regression & Survival Analysis
09	Transcriptome Analysis
10	Microbiome Analysis
11	Genome Analysis
12	Single-cell RNA-seq Analysis
13	Spatial Transcriptomics
14	Network Analysis
15	Multi-omics Integration & Case Studies
16	Other Omics & Course Wrap-up

Talk 07 — Statistical Analysis & Machine Learning Basics

- Descriptive statistics & summary tables
- Hypothesis testing: t-test, Wilcoxon, ANOVA, Kruskal-Wallis,

Additional Omics Topics (for future expansion)

- Drug repurposing & target prediction
- Pharmacogenomics
- Metagenomic binning & MAG analysis
- CRISPR screening data analysis
- Immuno-oncology (TCR/BCR repertoire)
- Long-read sequencing (Nanopore, PacBio) data analysis

Last updated: 2026-05-27 — Added Course Overview + R History to Talk 01; expanded Part 2 to 10 talks